

Lab 3: Component Modelling & Architectural Pattern Selection

Centralized Audit Trail Compliance Engine

Selected Architectural Pattern: Microservices Architecture

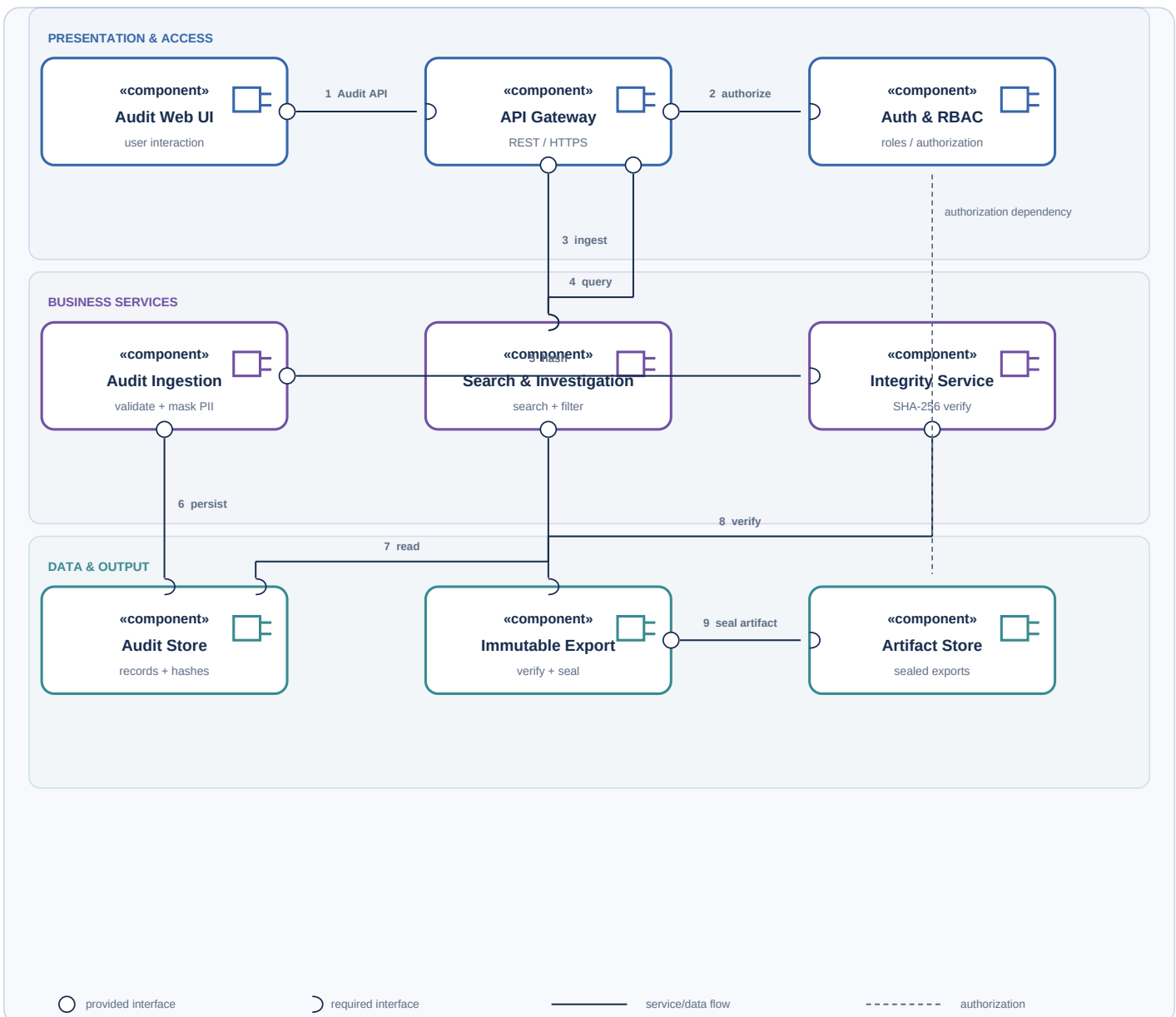
1. System Overview

The Centralized Audit Trail Compliance Engine ingests enterprise audit events, validates and protects records, masks sensitive PII, stores integrity hashes, supports authorized investigation, and produces verifiable immutable exports. The component model separates these responsibilities into independently deployable service boundaries.

2. Main Components

Component	Responsibility
Audit Web UI	User-facing search, filtering, integrity-status and export interactions.
API Gateway	Controlled REST/HTTPS entry point; routes requests to protected services.
Auth & RBAC	Authentication and role-based authorization for protected operations.
Audit Ingestion	Receives structured audit events; validates fields and masks sensitive PII.
Search & Investigation	Historical search, multi-criteria filtering and integrity-status viewing.
Integrity Service	Calculates and verifies SHA-256 integrity values.
Immutable Export	Retrieves evidence, verifies integrity, generates and seals export artifacts.
Audit Store	Persistent storage for protected audit records and integrity hashes.
Artifact Store	Persistent storage for sealed export artifacts and verification metadata.

3. UML Component Diagram



4. Interface and Data-Flow Details

Interface	Connected Components	Purpose
Audit API	Audit Web UI ↔ API Gateway	Controlled REST/HTTPS entry point for user operations.
Ingest API	API Gateway → Audit Ingestion	Submit structured JSON audit events for validation and protection.
Search API	API Gateway → Search & Investigation	Search and filter historical audit records.
RBAC / Authorization	Auth & RBAC → protected services	Authorize access before protected audit operations.
Hash / Verify	Audit Ingestion ↔ Integrity Service; Integrity Service ↔ Export	Calculate and verify SHA-256 integrity values.
Persist / Query	Audit Ingestion / Search ↔ Audit Store	Persist protected records and query historical evidence.
Export API	Search & Investigation ↔ Immutable Export	Request selected evidence for export generation.
Seal Artifact	Immutable Export → Artifact Store	Store the sealed immutable export and verification metadata.

5. Architecture Selection & Written Justification

Architecture Selection: We chose Microservices Architecture for the Centralized Audit Trail Compliance Engine.

Architectural Choice: The system is divided into independently deployable service boundaries for ingestion, investigation, integrity verification, immutable export, and access control, with protected audit and artifact storage.

Reason 1 — Functional separation: Lab 1 and Lab 2 define distinct capabilities for ingestion, PII protection, search/filtering, integrity, export, and access control. Separating these capabilities keeps responsibilities focused and permits performance-sensitive services to scale independently.

Reason 2 — Alignment with the backlog: Lab 2 contains four epics: Audit Log Ingestion & Protection, Audit Search & Investigation, Immutable Audit Export, and Security & Access Control. These boundaries map naturally to service responsibilities in the component model.

Security Advantage: Auth & RBAC can enforce authorization before protected audit operations. PII masking occurs in the ingestion path, SHA-256 verification provides an integrity check, and export generation can require successful verification before sealing.

Performance Benefit: Lab 1 specifies complex filtering across up to 10 million historical records with a target response time below one second under the defined benchmark workload. Separating Search & Investigation provides an architectural path to scale query resources independently of ingestion and export processing.

6. Component Interaction Flow

A typical audit flow begins at the Audit Web UI and passes through the API Gateway. Auth & RBAC authorizes protected operations. Audit events are processed by Audit Ingestion, where required fields are validated and sensitive PII is masked before persistence. The Integrity Service calculates or verifies SHA-256 values. Authorized users can search and filter historical records through Search & Investigation. For an export, selected evidence is verified before Immutable Export generates and seals the artifact in Artifact Store.

The microservices boundaries make security-sensitive operations explicit, isolate the query workload, and trace directly to the requirements and completed Lab 2 backlog. The trade-off is additional service-to-service communication, deployment, observability, and data-consistency complexity.

Lab 1 → Lab 2 → Lab 3 Traceability

FR-001/002 → Ingestion & protection | FR-003 → Integrity Service + Audit Store | FR-004 → Search & Investigation
FR-005 → Immutable Export + Artifact Store | NFR security/privacy → Auth & RBAC | NFR performance → scalable search path